CM4Stack

SKU:K127/K127-US/K127-EU





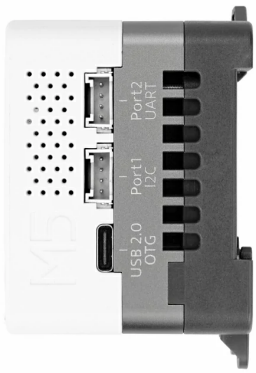


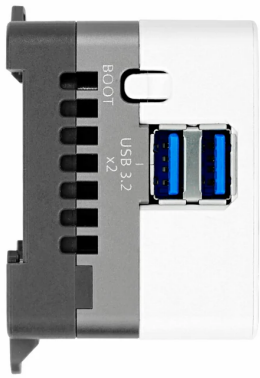










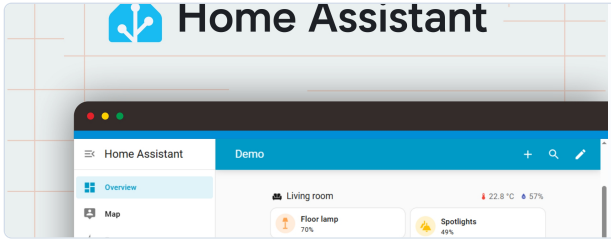




Description

CM4Stack is an all-in-one micro host based on the **Raspberry Pi CM4 core board**, using the CM4104032, with built-in Wi-Fi, 4G memory, and 32Gb eMMC. The host includes a variety of peripherals: a 2.0-inch IPS-LCD screen with capacitive **touch**, resolution 320x240, 2 USB3.2-A ports (PCIe + ASM3042), a USB-C (2.0) interface supporting **OTG** and host power supply, 1 high-resolution multimedia signal interface, 1 Gigabit Ethernet interface (RJ45), 1 DC-JACK power input interface (12V@3A); 2 HY2.0-4P expansion ports, which are I2C and UART interfaces, built-in 2W speaker with AW88298 16-Bits I2S amplifier, 1 security encryption chip ATECC608B, 1 RTC chip BM8563, built-in active cooling fan and aluminum alloy heat dissipation frame. This product structurally supports DIN rail, wall mount, and screw fixing methods, suitable for industrial automation, smart home, IoT, and other fields.

Tutorial



Home Assistant

This tutorial will introduce you to burning the Home Assistant image to CM4Stack



Raspberrypi OS

This tutorial will introduce you to burning the Raspberrypi OS image to CM4Stack



CM4Stack QuickStart

CM4Stack startup configuration and related tutorials

SKU	Difference
K127	Comes with Japan power adapter
K127-US	Comes with US power adapter
K127-EU	Comes with EU power adapter

Note

When connecting to CM4Stack, it is recommended to use a high-quality USB cable certified by USB-IF to prevent device malfunction.

Features

- High-performance embedded system: Built-in powerful chips and interfaces, equipped with a 2.0-inch LCD screen with touch functionality, capable of displaying high-quality video and images.
- Multiple interfaces and functions: Includes antenna interface, 2W speaker, high-resolution multimedia signal interface, USB-C (2.0) interface, USB3.2 interface, Gigabit Ethernet interface (RJ45), I2C interface, serial port, and DC power input port, supporting various peripherals and data transmission methods.
- Security features: Equipped with a security authentication chip AW32901ID-KEY (ATECC608B) for secure data transmission and storage.
- Clock function: Equipped with RTC chip BM8563 for clock functionality.
- Cooling design: Equipped with a cooling base and fan to ensure long-term stable operation of the system.

Includes

- 1 x CM4Stack
- 1 x CM4Stack Power Adapter (input: 100-240VAC, OUTPUT: DC_12V 3A)
- 1 x Hex Key 2mm
- 1 x CM4Stack Manual
- 1 x CM4Stack Base Clip
- 1 x CM4Stack Screw Fixing Kit

Applications

- Industrial Automation
- Smart Home
- Internet of Things (IoT)
- Education and Development

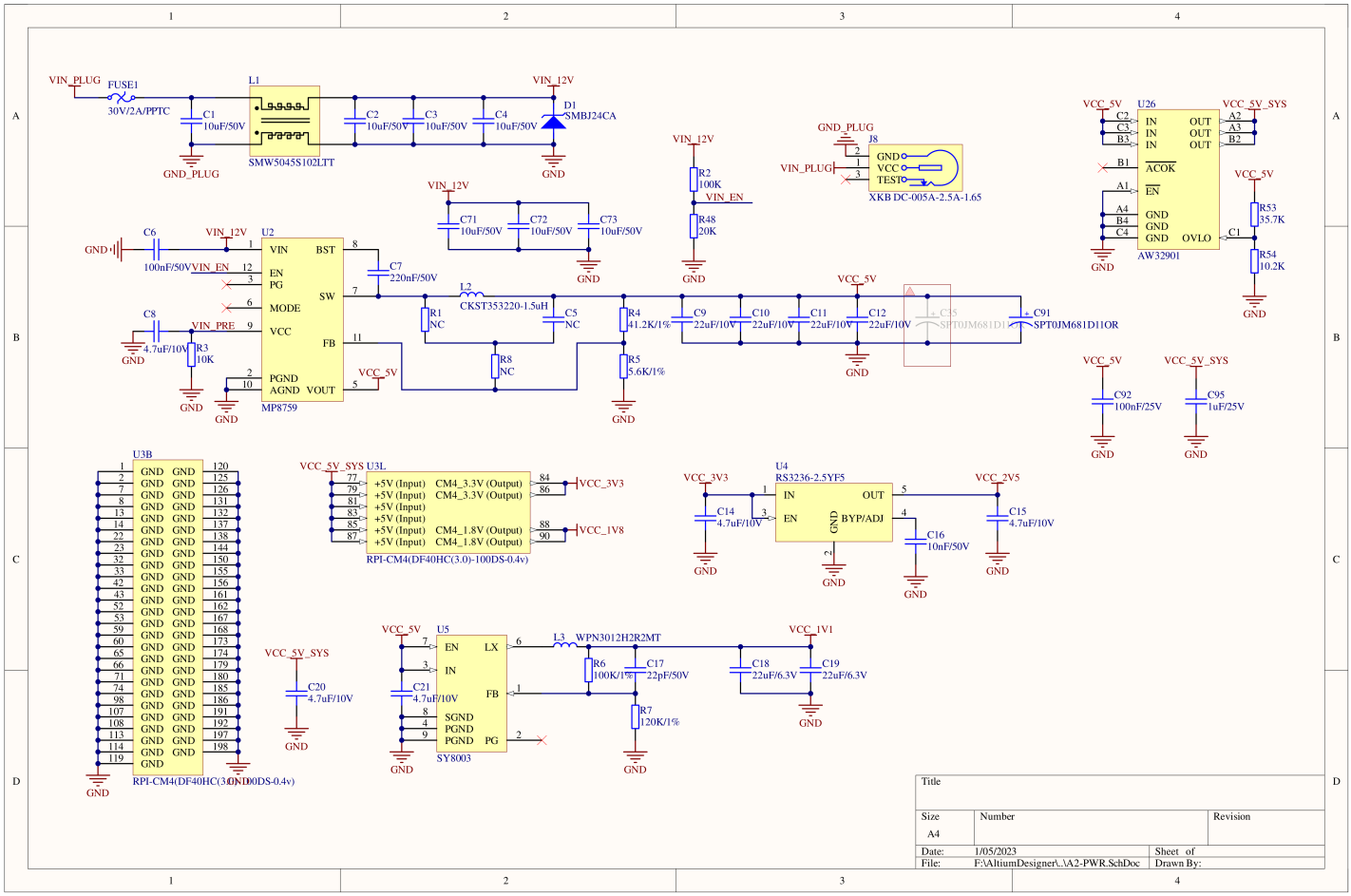
Specifications

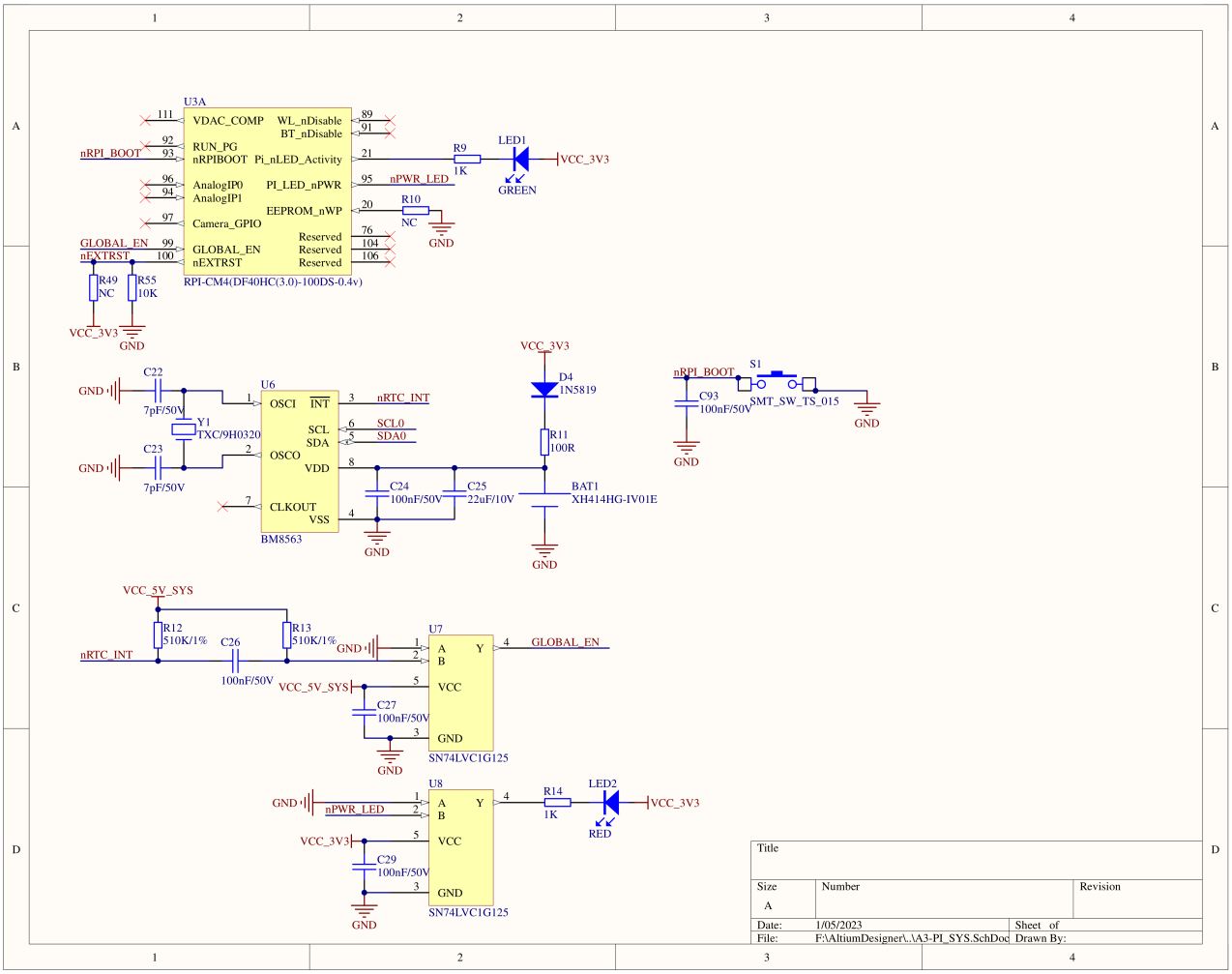
Specification	Parameter
Module Model	CM4104032
SoC	Broadcom BCM2711@Quad-core Cortex-A72 64-bit, up to 1.5GHz
RAM	4GB
Flash	32GB eMMC
2.0 inch LCD	Screen with touch functionality and ST7789V2 controller
Touch Chip Solution	GT911
Antenna Interface	For connecting external antennas
Speaker	2W audio output with AW88298 driver
USB-C Interface	USB 2.0 interface with OTG functionality, powered by MP8759 and AW32901 controllers
I2C Interface	For connecting I2C devices
Serial Port	Communication interface for transmitting and receiving serial data
AC Power Input Port	Supports 100-240VAC AC power input

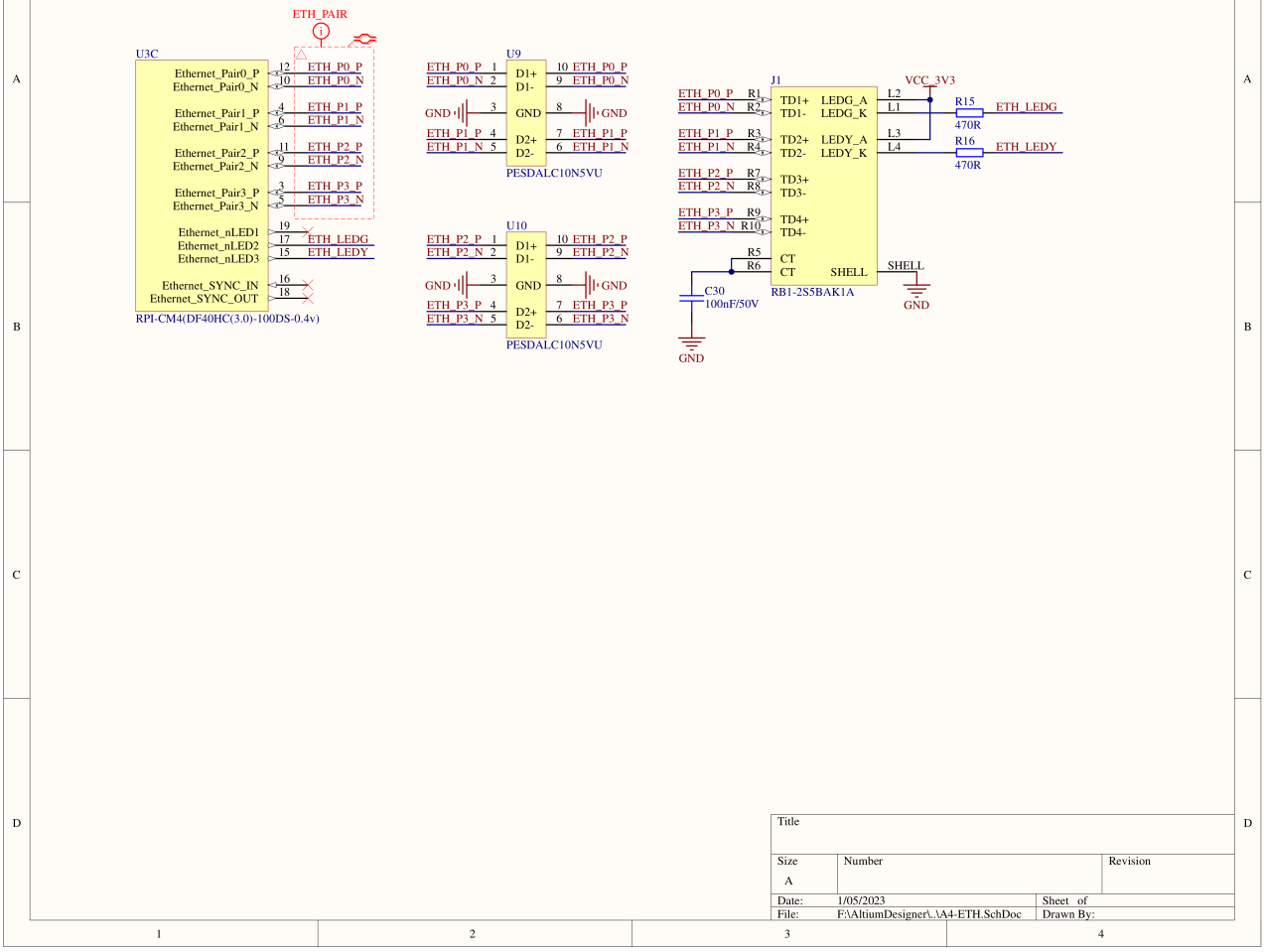
Specification	Parameter
DC Power Input Port	Supports 12V/3A DC power input
Gigabit Ethernet Interface	RJ45 interface for connecting to Gigabit Ethernet network, speed: 1Gbps
USB3.2 Interface	Supports USB 3.2 protocol, uses ASM3042 chip, high-speed PCIe bridge
Cooling Base	Made of aluminum alloy, suitable for DIN rail installation
Cooling Fan	For heat dissipation, uses ME1502 chip
Download Button	Button for entering download mode
High-resolution Multimedia Signal Interface	Interface for transmitting high-definition video and audio
RTC	Real-time clock module with BM8563
Chip ID-KEY	For device security authentication, uses ATECC608B chip
Aluminum Alloy Cooling Base	DIN rail installation cooling base with built-in cooling fan (ME1502)
SY8003 and ME1502	Power management and control integrated circuits
AW88298	Audio amplifier integrated circuit
MP8759 and AW32901	USB power delivery and Type-C interface control integrated circuits
ID-KEY (ATECC608B)	Encryption authentication integrated circuit for secure communication
ASM3042	USB 3.2 main controller
BM8563	Real-time clock integrated circuit
WIFI	2.4 GHz, 5.0 GHz IEEE 802.11 b/g/n/ac
Product Size	59 x 46 x 44mm
Product Weight	116.4g
Package Size	160 x 107 x 47mm
Gross Weight	330.5g

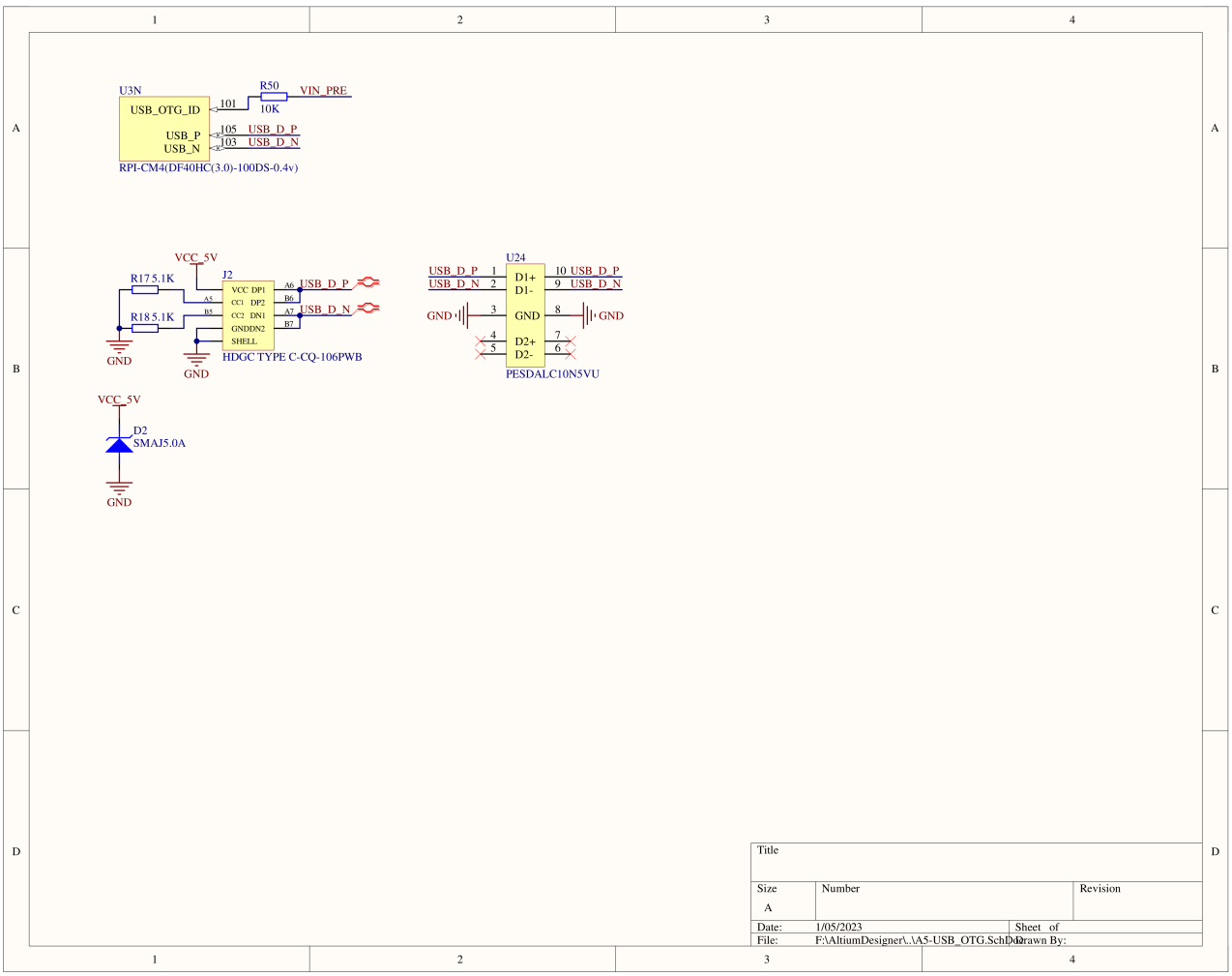
Schematics

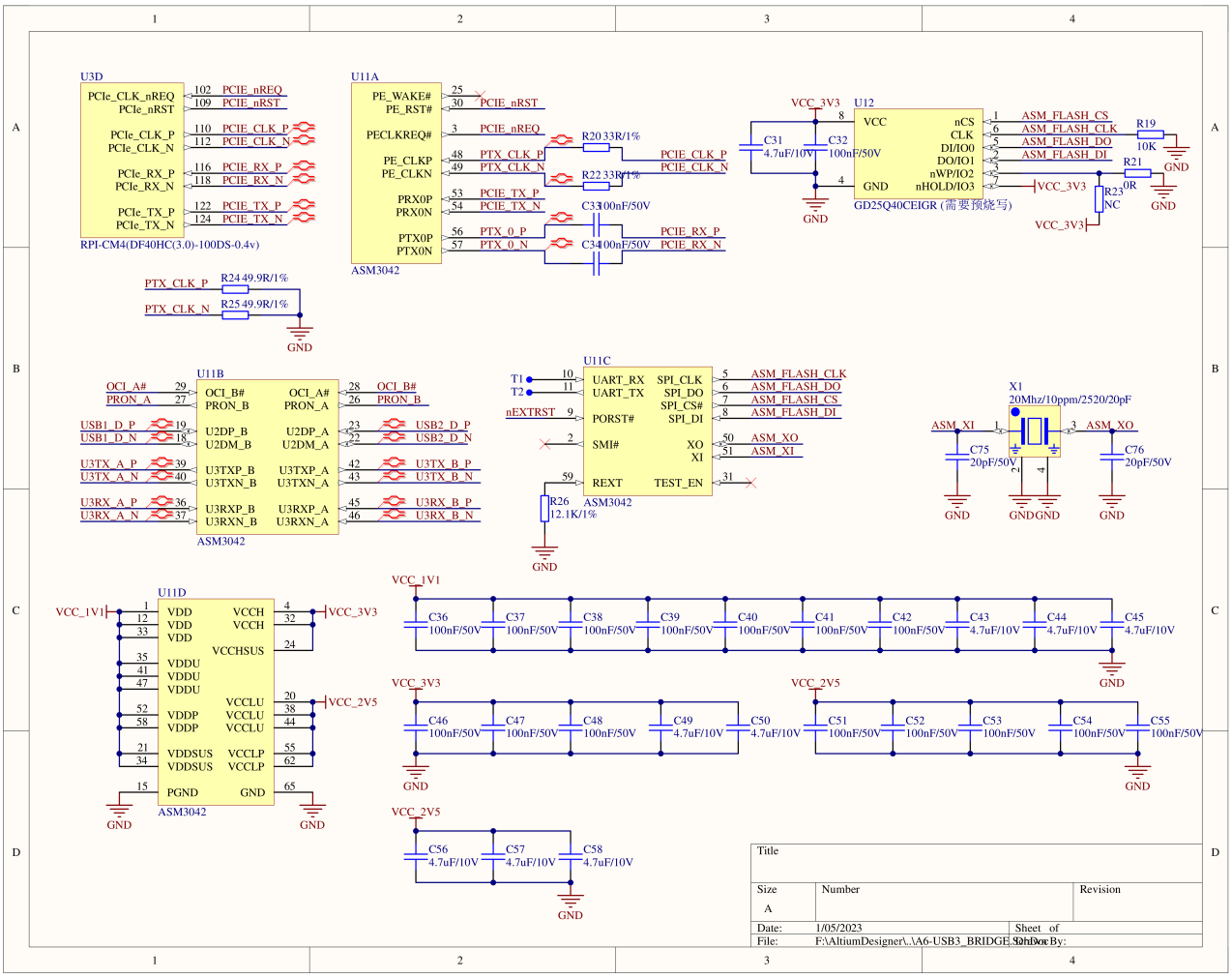
- CM4Stack Schematics PDF

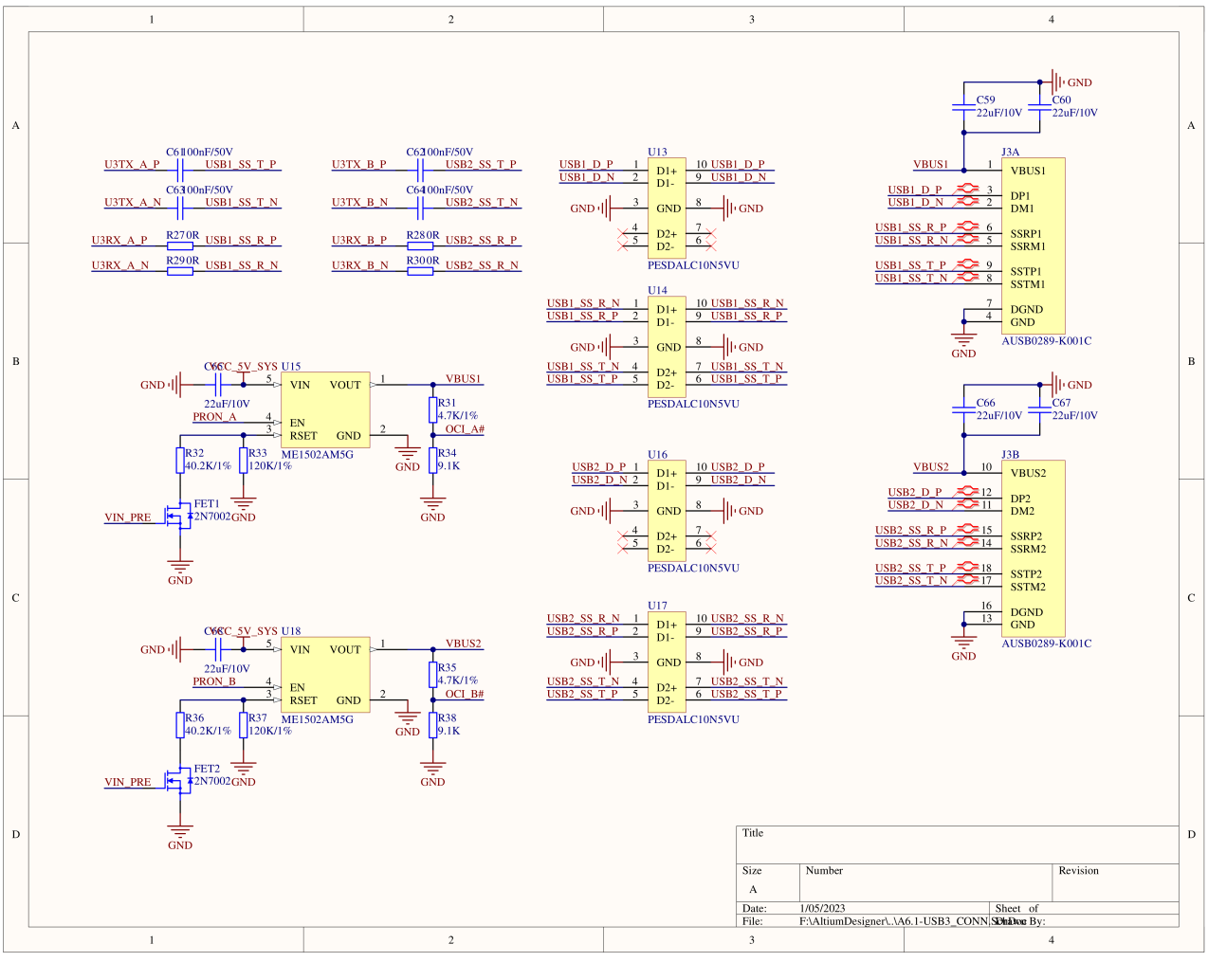

VIN_PLUG
FUSE1
30V/2A/PPTC
L1
SMW5045S102LTT
D1
SMBJ24CA
VIN_12V
GND_PLUG
GND
C1 10uF/50V
C2 10uF/50V
C3 10uF/50V
C4 10uF/50V
C71 10uF/50V
C72 10uF/50V
C73 10uF/50V
R2 100K
R48 20K
VIN_EN
GND_PLUG
J8
XKB DC-005A-2.5A-1.65
U26
AW32901
VCC_5V
VCC_5V_SYS
R53 35.7K
R54 10.2K
C6 100nF/50V
C8 4.7uF/10V
R3 10K
VIN_PRE
U2
MP8759
VIN
BST
EN
PG
SW
MODE
VCC
FB
PGND
AGND
VOUT
C7 220nF/50V
L2 CKST353220-1.5uH
R1 NC
C5 NC
R8 NC
R4 41.2K/1%
R5 5.6K/1%
C9 22uF/10V
C10 22uF/10V
C11 22uF/10V
C12 22uF/10V
C35 SPT0JM681D1H
C91 SPT0JM681D1I1OR
C92 100nF/25V
C95 1uF/25V
U3B
RPI-CM4(DF40HC(3.0)-100DS-0.4v)
U3L
RPI-CM4(DF40HC(3.0)-100DS-0.4v)
+5V (Input)
CM4_3.3V (Output)
CM4_1.8V (Output)
VCC_3V3
VCC_1V8
U4
RS3236-2.5YF5
IN
OUT
EN
GND
BYP/ADJ
VCC_3V3
VCC_2V5
C14 4.7uF/10V
C15 4.7uF/10V
C16 10nF/50V
U5
SY8003
EN
LX
IN
FB
SGND
PGND
PG
L3 WPN3012H2R2MT
R6 100K/1%
R7 120K/1%
C17 22pF/50V
C18 22uF/6.3V
C19 22uF/6.3V
VCC_1V1
C20 4.7uF/10V
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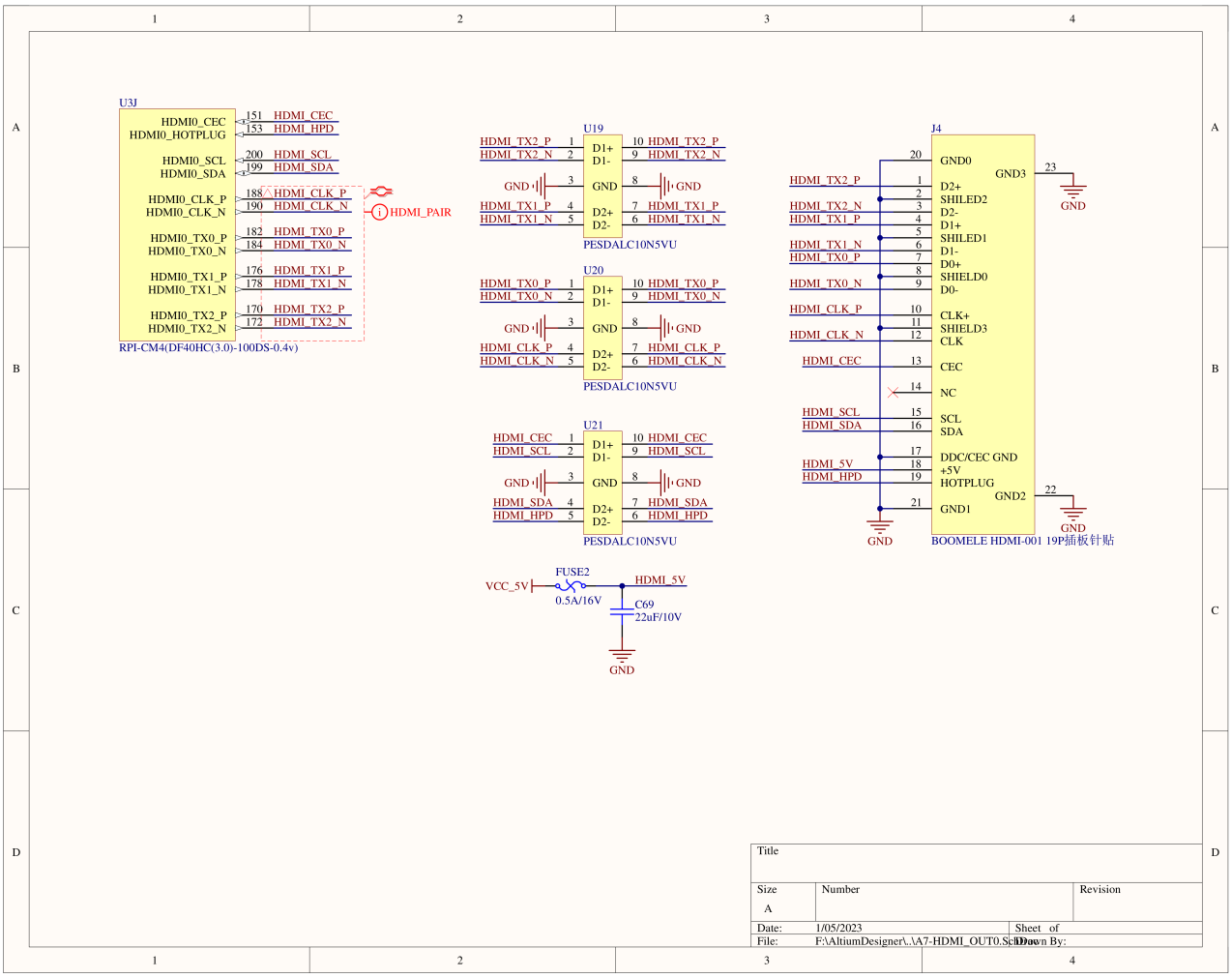

U3A
RPI-CM4(DF40HC(3.0)-100DS-0.4v)
VDAC_COMP
RUN_PG
nRPIBOOT
AnalogIP0
AnalogIP1
Camera_GPIO
GLOBAL_EN
nEXTRST
WL_nDisable
BT_nDisable
Pi_nLED_Activity
PI_LED_nPWR
EEPROM_nWP
Reserved
nRPI_BOOT
nPWR_LED
LED1
GREEN
R9 1K
R10 NC
R49 NC
R55 10K
VCC_3V3
U6
BM8563
OSCI
OSCO
INT
SCL
SDA
VDD
CLKOUT
VSS
nRTC_INT
SCL0
SDA0
C22 7pF/50V
C23 7pF/50V
Y1 TXC/9H0320
C24 100nF/50V
C25 22uF/10V
D4 1N5819
R11 100R
BAT1 XH414HG-IV01E
S1 SMT_SW_TS_015
C93 100nF/50V
U7
U8
SN74LVC1G125
A
B
Y
VCC
GND
GLOBAL_EN
R12 510K/1%
R13 510K/1%
C26 100nF/50V
C27 100nF/50V
R14 1K
LED2
RED
C29 100nF/50V
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Revision
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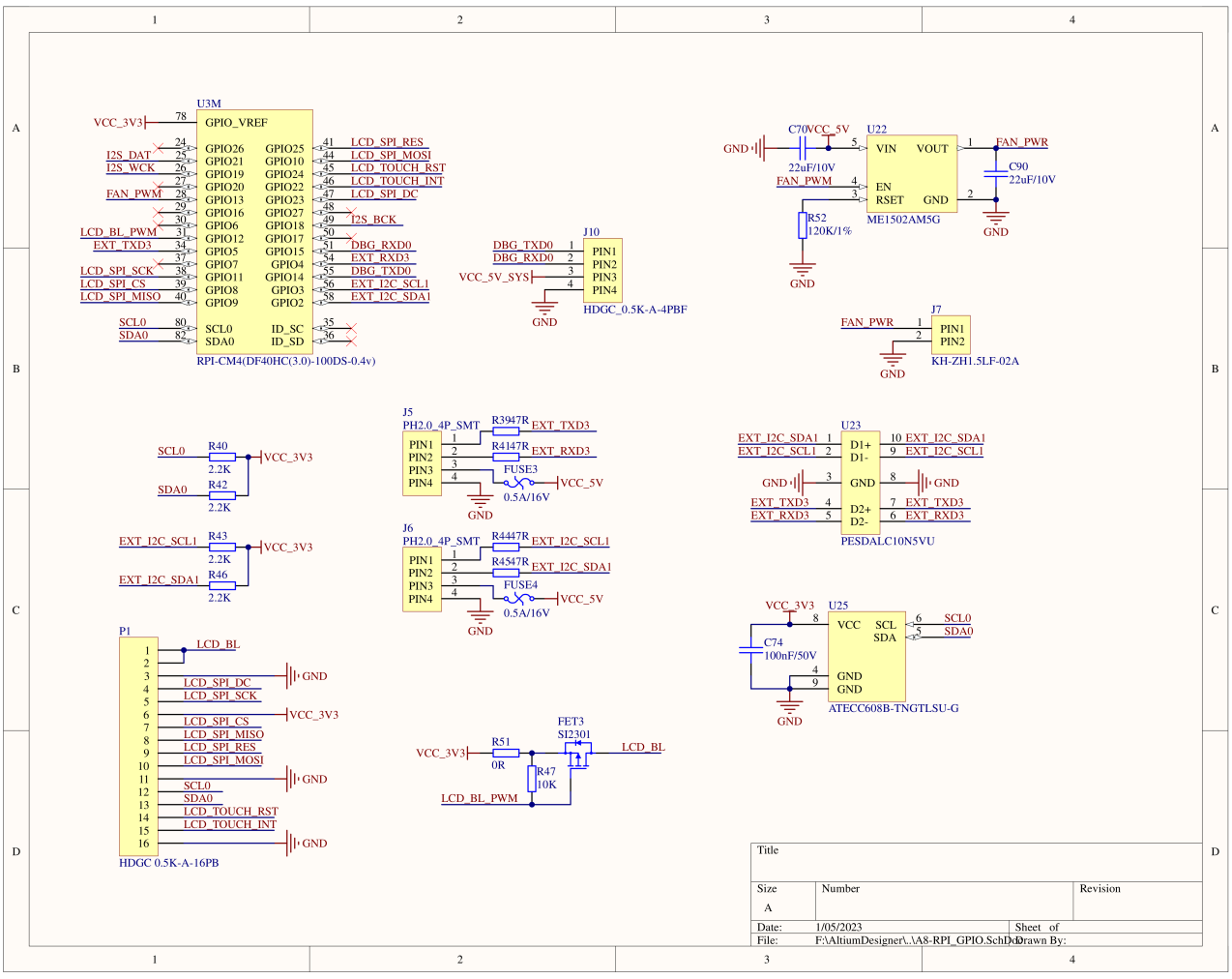

ETH_PAIR
U3C
Ethernet_Pair0_P
Ethernet_Pair0_N
Ethernet_Pair1_P
Ethernet_Pair1_N
Ethernet_Pair2_P
Ethernet_Pair2_N
Ethernet_Pair3_P
Ethernet_Pair3_N
Ethernet_nLED1
Ethernet_nLED2
Ethernet_nLED3
Ethernet_SYNC_IN
Ethernet_SYNC_OUT
ETH_P0_P
ETH_P0_N
ETH_P1_P
ETH_P1_N
ETH_P2_P
ETH_P2_N
ETH_P3_P
ETH_P3_N
ETH_LEDG
ETH_LEDY
RPI-CM4(DF40HC(3.0)-100DS-0.4v)
U9
D1+
D1-
GND
D2+
D2-
PESDALC10N5VU
U10
J1
TD1+
TD1-
TD2+
TD2-
TD3+
TD3-
TD4+
TD4-
CT
SHELL
LEDG_A
LEDG_K
LEDY_A
LEDY_K
VCC_3V3
R15
R16
470R
C30
100nF/50V
RB1-2S5BAK1A
Title
Size
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Number
Revision
Date: 1/05/2023
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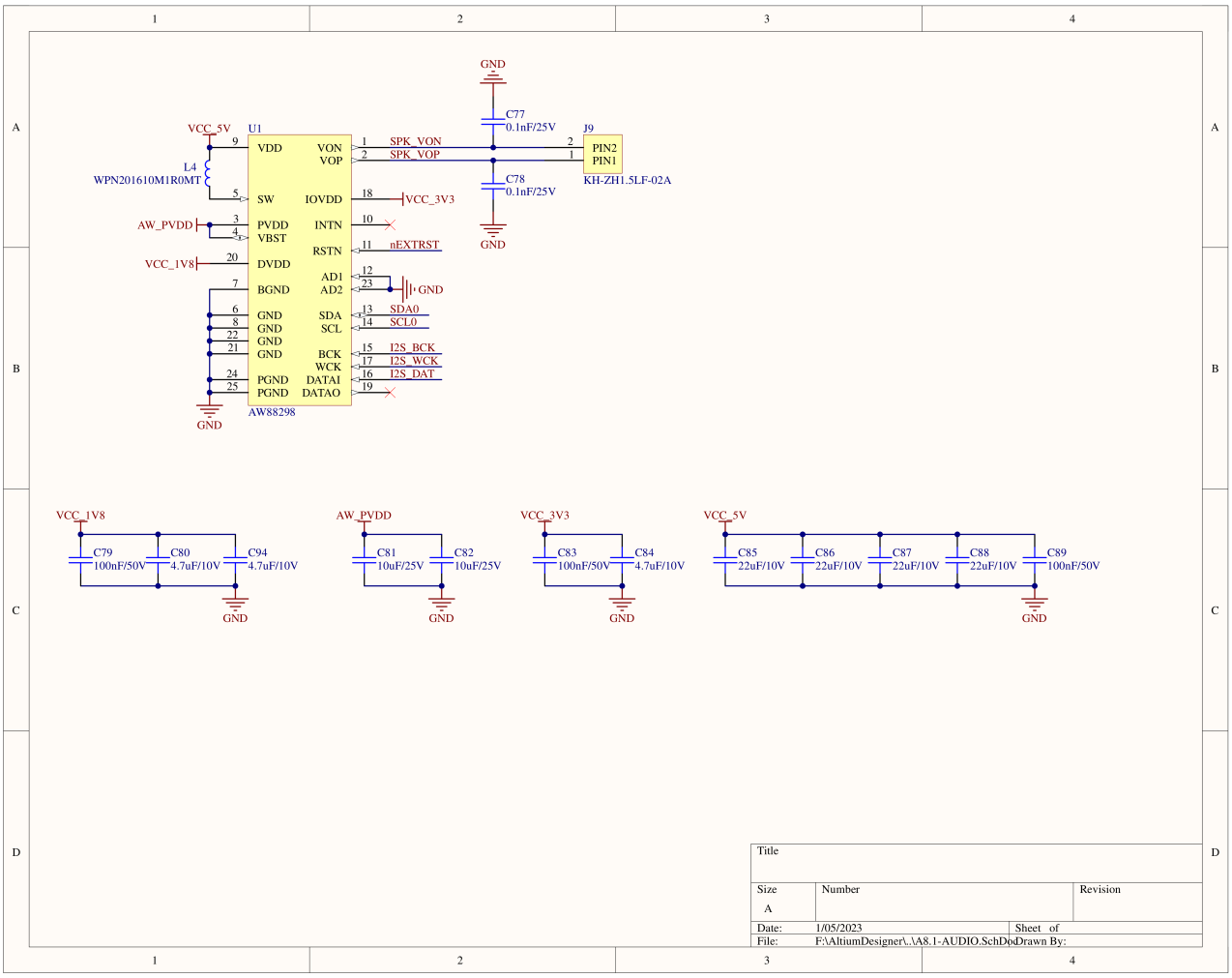

U3N
USB_OTG_ID
USB_P
USB_N
USB_D_P
USB_D_N
R50
10K
VIN_PRE
RPI-CM4(DF40HC(3.0)-100DS-0.4v)
VCC_5V
R17 5.1K
R18 5.1K
J2
HDGC TYPE C-CQ-106PWB
U24
PESDALC10N5VU
D2
SMAJ5.0A
Title
Size
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Number
Revision
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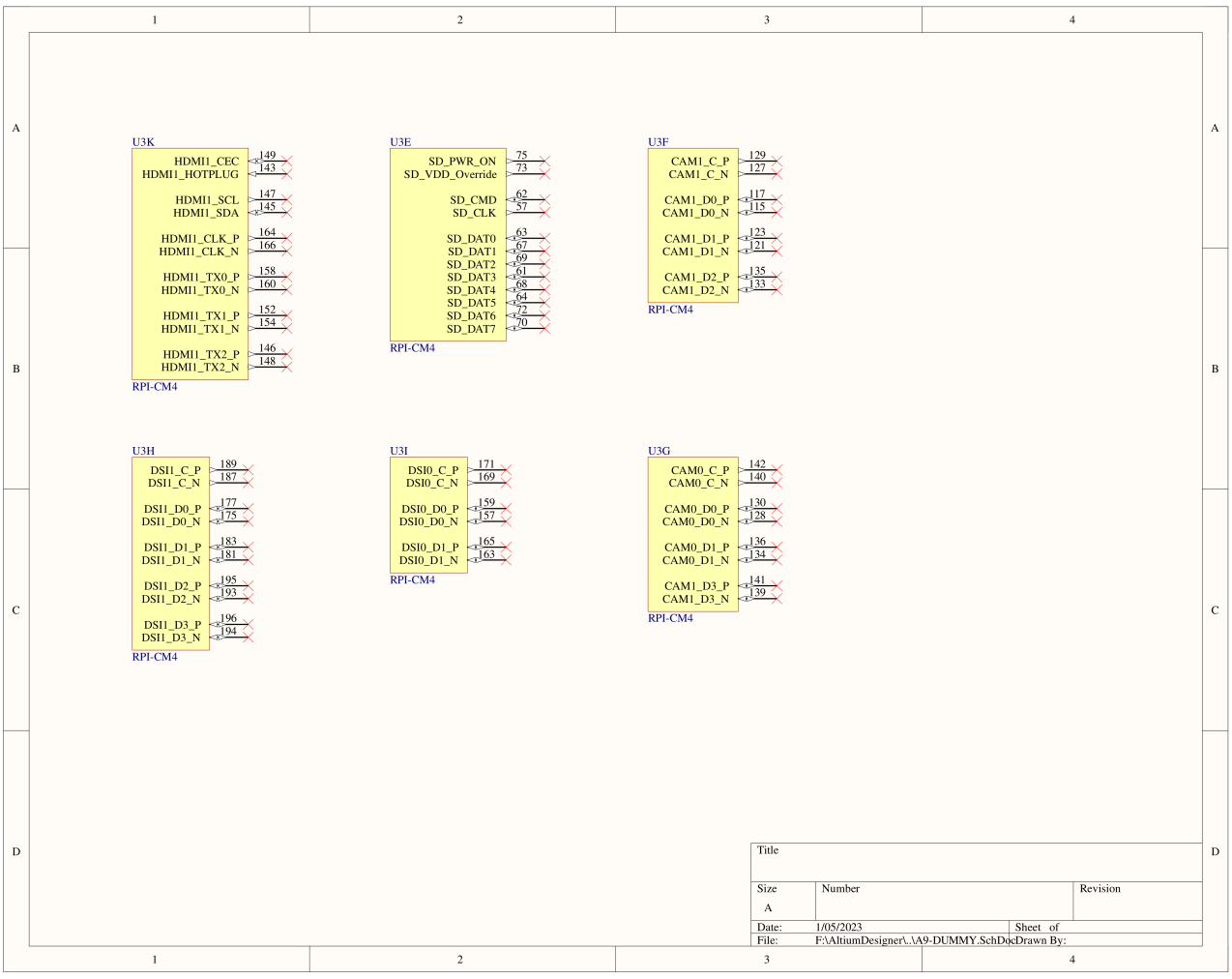

U3D
RPI-CM4(DF40HC(3.0)-100DS-0.4v)
U11A
ASM3042
U12
GD25Q40CEIGR (需要预烧写)
U11B
ASM3042
U11C
ASM3042
X1
20Mhz/10ppm/2520/20pF
U11D
ASM3042
Title
Size
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Number
Revision
Date: 1/05/2023
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U13
PESDALC10N5VU
U14
PESDALC10N5VU
U16
PESDALC10N5VU
U17
PESDALC10N5VU
U15
ME1502AM5G
U18
ME1502AM5G
J3A
AUSB0289-K001C
J3B
AUSB0289-K001C
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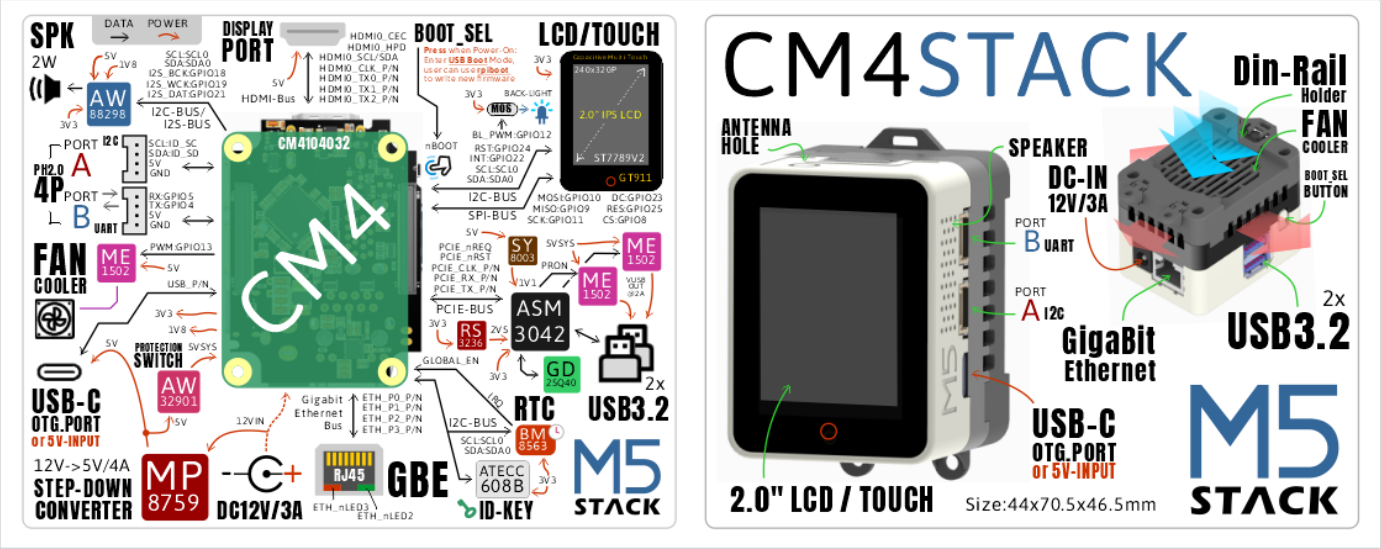

U3J
HDMI0_CEC
HDMI0_HOTPLUG
HDMI0_SCL
HDMI0_SDA
HDMI0_CLK_P
HDMI0_CLK_N
HDMI0_TX0_P
HDMI0_TX0_N
HDMI0_TX1_P
HDMI0_TX1_N
HDMI0_TX2_P
HDMI0_TX2_N
151 HDMI_CEC
153 HDMI_HPD
200 HDMI_SCL
199 HDMI_SDA
188 HDMI_CLK_P
190 HDMI_CLK_N
182 HDMI_TX0_P
184 HDMI_TX0_N
176 HDMI_TX1_P
178 HDMI_TX1_N
170 HDMI_TX2_P
172 HDMI_TX2_N
HDMI_PAIR
RPI-CM4(DF40HC(3.0)-100DS-0.4v)
U19
HDMI_TX2_P 1
HDMI_TX2_N 2
GND 3
HDMI_TX1_P 4
HDMI_TX1_N 5
D1+
D1-
GND
D2+
D2-
10 HDMI_TX2_P
9 HDMI_TX2_N
8 GND
7 HDMI_TX1_P
6 HDMI_TX1_N
PESDALC10N5VU
U20
HDMI_TX0_P 1
HDMI_TX0_N 2
GND 3
HDMI_CLK_P 4
HDMI_CLK_N 5
10 HDMI_TX0_P
9 HDMI_TX0_N
8 GND
7 HDMI_CLK_P
6 HDMI_CLK_N
PESDALC10N5VU
U21
HDMI_CEC 1
HDMI_SCL 2
GND 3
HDMI_SDA 4
HDMI_HPD 5
10 HDMI_CEC
9 HDMI_SCL
8 GND
7 HDMI_SDA
6 HDMI_HPD
PESDALC10N5VU
FUSE2
VCC_5V
0.5A/16V
HDMI_5V
C69
22uF/10V
GND
J4
HDMI_TX2_P
HDMI_TX2_N
HDMI_TX1_P
HDMI_TX1_N
HDMI_TX0_P
HDMI_TX0_N
HDMI_CLK_P
HDMI_CLK_N
HDMI_CEC
HDMI_SCL
HDMI_SDA
HDMI_5V
HDMI_HPD
GND0
D2+
SHILED2
D2-
D1+
SHILED1
D1-
D0+
SHIELD0
D0-
CLK+
SHIELD3
CLK
CEC
NC
SCL
SDA
DDC/CEC GND
+5V
HOTPLUG
GND1
GND2
GND3
BOOMELE HDMI-001 19P插板针贴
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Revision
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U3M
VCC_3V3
GPIO_VREF
I2S_DAT
I2S_WCK
FAN_PWM
LCD_BL_PWM
EXT_TXD3
LCD_SPI_SCK
LCD_SPI_CS
LCD_SPI_MISO
SCL0
SDA0
LCD_SPI_RES
LCD_SPI_MOSI
LCD_TOUCH_RST
LCD_TOUCH_INT
LCD_SPI_DC
I2S_BCK
DBG_RXD0
EXT_RXD3
DBG_TXD0
EXT_I2C_SCL1
EXT_I2C_SDA1
ID_SC
ID_SD
RPI-CM4(DF40HC(3.0)-100DS-0.4v)
J10
DBG_TXD0
DBG_RXD0
VCC_5V_SYS
PIN1
PIN2
PIN3
PIN4
HDGC_0.5K-A-4PBF
C70VCC_5V
22uF/10V
FAN_PWM
R52
120K/1%
U22
VIN
VOUT
EN
RSET
GND
ME1502AM5G
FAN_PWR
C90
22uF/10V
J7
KH-ZH1.5LF-02A
SCL0
SDA0
R40
R42
2.2K
VCC_3V3
EXT_I2C_SCL1
EXT_I2C_SDA1
R43
R46
J5
PH2.0_4P_SMT
R3947R
EXT_TXD3
R4147R
EXT_RXD3
FUSE3
0.5A/16V
VCC_5V
J6
R4447R
EXT_I2C_SCL1
R4547R
EXT_I2C_SDA1
FUSE4
U23
EXT_I2C_SDA1
EXT_I2C_SCL1
EXT_TXD3
EXT_RXD3
PESDALC10N5VU
U25
VCC
SCL
SDA
C74
100nF/50V
ATECC608B-TNGTLSU-G
P1
LCD_BL
LCD_SPI_DC
LCD_SPI_SCK
LCD_SPI_CS
LCD_SPI_MISO
LCD_SPI_RES
LCD_SPI_MOSI
SCL0
SDA0
LCD_TOUCH_RST
LCD_TOUCH_INT
HDGC 0.5K-A-16PB
FET3
SI2301
R51
0R
R47
10K
LCD_BL_PWM
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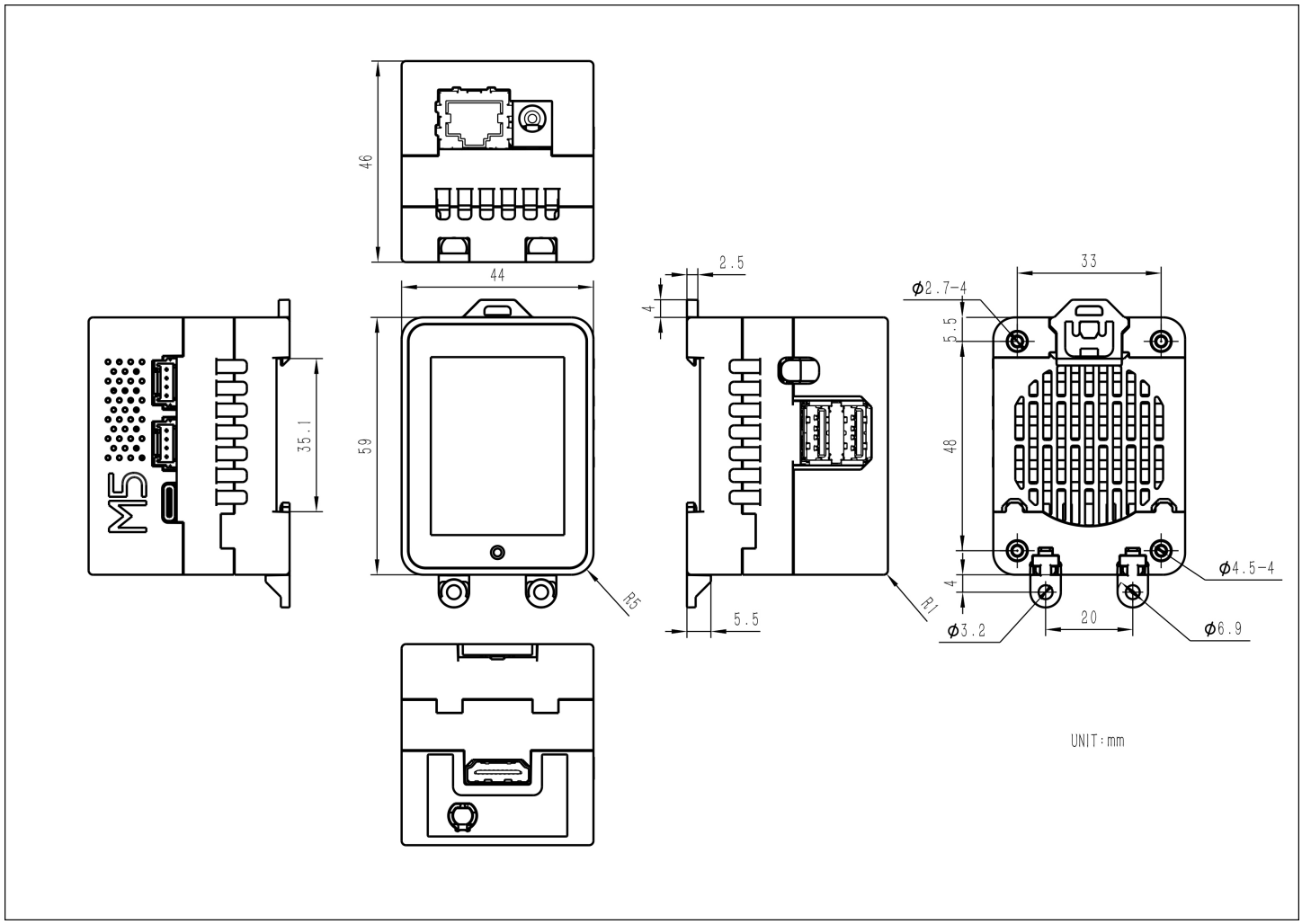

GND
C77
0.1nF/25V
J9
VCC_5V
U1
VDD
VON
VOP
SPK_VON
SPK_VOP
PIN2
PIN1
KH-ZH1.5LF-02A
L4
WPN201610M1R0MT
C78
0.1nF/25V
SW
IOVDD
VCC_3V3
AW_PVDD
PVDD
VBST
INTN
RSTN
nEXTRST
VCC_1V8
DVDD
AD1
AD2
BGND
SDA0
SCL0
SDA
SCL
BCK
WCK
DATAI
DATAO
I2S_BCK
I2S_WCK
I2S_DAT
AW88298
C79
100nF/50V
C80
4.7uF/10V
C94
4.7uF/10V
C81
10uF/25V
C82
10uF/25V
C83
100nF/50V
C84
4.7uF/10V
C85
22uF/10V
C86
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C87
22uF/10V
C88
22uF/10V
C89
100nF/50V
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Revision
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U3K
HDMI1_CEC
HDMI1_HOTPLUG
HDMI1_SCL
HDMI1_SDA
HDMI1_CLK_P
HDMI1_CLK_N
HDMI1_TX0_P
HDMI1_TX0_N
HDMI1_TX1_P
HDMI1_TX1_N
HDMI1_TX2_P
HDMI1_TX2_N
RPI-CM4
U3E
SD_PWR_ON
SD_VDD_Override
SD_CMD
SD_CLK
SD_DAT0
SD_DAT1
SD_DAT2
SD_DAT3
SD_DAT4
SD_DAT5
SD_DAT6
SD_DAT7
RPI-CM4
U3F
CAM1_C_P
CAM1_C_N
CAM1_D0_P
CAM1_D0_N
CAM1_D1_P
CAM1_D1_N
CAM1_D2_P
CAM1_D2_N
RPI-CM4
U3H
DSI1_C_P
DSI1_C_N
DSI1_D0_P
DSI1_D0_N
DSI1_D1_P
DSI1_D1_N
DSI1_D2_P
DSI1_D2_N
DSI1_D3_P
DSI1_D3_N
RPI-CM4
U3I
DSI0_C_P
DSI0_C_N
DSI0_D0_P
DSI0_D0_N
DSI0_D1_P
DSI0_D1_N
RPI-CM4
U3G
CAM0_C_P
CAM0_C_N
CAM0_D0_P
CAM0_D0_N
CAM0_D1_P
CAM0_D1_N
CAM1_D3_P
CAM1_D3_N
RPI-CM4
Title
Size
A
Number
Revision
Date: 1/05/2023
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File: F:\AltiumDesigner\..\A9-DUMMY.SchDoc
Drawn By:

PinMap



Model Size



Structure

- CM4Stack Structure Files

Datasheets

- CM4Stack Quick Guide
- CM4-datasheet
- CM4IO-datasheet
- ST7789V2
- AW88298
- MP8759
- AW32901
- SY8003
- ME1502
- BM8563
- ATECC608B
- AC adapter

Softwares

Quick Start

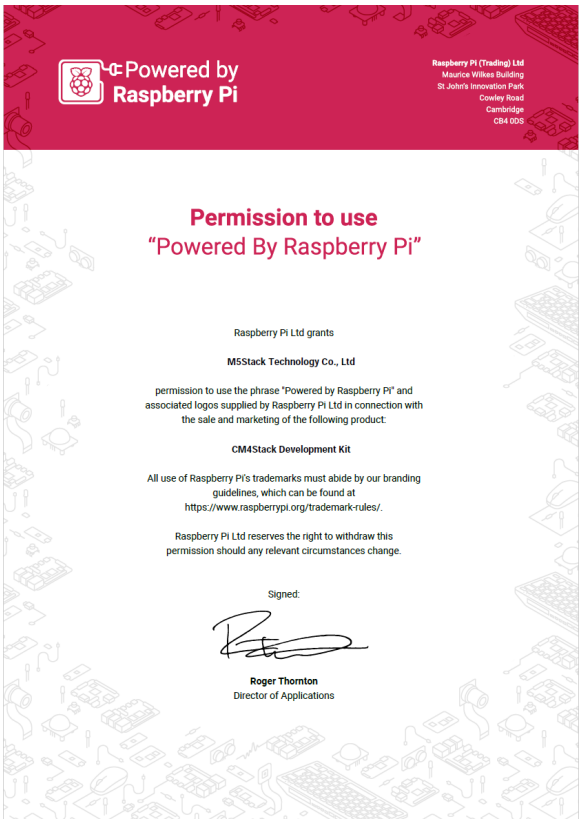
- CM4Stack HAOS Flashing Tutorial
- Flashing Raspberry Pi OS Image
- CM4Stack Boot Configuration and Related Tutorials

Other

- M5stack-Linux-bsp Clock Demo
- CM4Stack_lvgl Demo

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